Evaluation of Strategy: Theory and Models

Richard P. Rumelt

Harvard Business School, 1978

This working paper was subsequently published as:

Rumelt, Richard P. "Evaluation of Strategy: Theory and Models." In *Strategic Management: A New View of Business Policy and Planning*, edited by Dan E. Schendel and Charles H. Hofer, pp. 196–216. Boston: Little, Brown, 1979.

The formulation of organizational strategy, when it occurs, is problem solving of the most unstructured sort. What little is known about how ill-structured problems are, or should be, dealt with suggests a three-part cycling process among: (1) problem structuring activities, (2) the generation of tentative solutions, and (3) the testing or evaluation of proposed solutions. The creative phases of this process draw heavily on perceptual skills and imagination. Here, pattern recognition and the ability to perceive new meanings amidst complexity play key roles. By contrast, it is the evaluation phase that presents the greatest opportunity to employ rational analysis. The situation is directly analogous to the scientific process wherein hypothesis generation remains a high art; the full power of the scientific method comes into play only after a clear hypothesis has been formulated.

If "evaluation" is the phase of strategy-making that is most amenable to analytic thought, what is the current state of the art? To treat this question, consider the following idealized problem: suppose one is given a reasonably comprehensive description of an organization, its environment and a proposed strategy. What are the legitimate grounds for evaluating the strategy and to what theories, knowledge, or models can one turn for help in making such an evaluation? The purpose of this article is to examine these questions in the light of recent research and conceptual contributions.

The idealized problem posed is somewhat artificial in that descriptions of strategy, organization, and environment are rarely "comprehensive" and strategy evaluation is more often an organizational process than an analytical task. But our interpretation of "state of the art" is that it implies the "best" that can be done rather than what some do, what might be done, or what is most frequently done. Accordingly, the primary concern will be with the limits of rational evaluation - with identifying the cut- ting edge between what can be said with reasonable surety about general classes of situations and those aspects of evaluation that remain heavily dependent upon intuition,

insight, and sharply situational knowledge. Strategy is a strongly contextual concept. At the core it deals with the adjustment of specific policies to particular situations. In looking at strategy evaluation, therefore, it will be helpful to associate methods of analysis with their appropriate contexts. We will begin with a discussion of the strategy concept itself, establish the functions it is supposed to perform, and identify those evaluation criteria that are context free - that are always valid. Central to this discussion will be the distinction between a solution to a problem and solving the right problem. We introduce the idea of "frame" as a way of speaking more precisely about this issue and indicate the special characteristics of "frame theories" that are useful in strategy evaluation.

The broadest context in general use is that of "rivalry." In discussing the context of rivalry we will show that many of the traditional tests for evaluating strategy are logical consequences of the nature of the strategy concept and the assumption of rivalry.

Armed with these observations and tools, a variety of models and theories will be discussed. In each case, the relevant context will be identified, the frame content examined, and the specific "tactical" content of the approach noted. After summarizing our analysis we will point out what seem to be the most fruitful directions for further research.

The Strategy Concept

To evaluate an entity is to determine its utility, truth, or efficacy. This, in turn, requires an appreciation of the function the entity is supposed to perform. Therefore, it will be helpful to examine the special characteristics of the concept of strategy and the role it is supposed to play in the guidance of organizations. Only then can the legitimate grounds for its appraisal be established.

The term "strategy" has a range of related meanings and authors have generally felt quite free to use it quite idiosyncratically. For game theorists, strategies are concrete actions or rules for choosing actions in a conflict situation; for some strategy is "high-level" or "long-term" planning, while others see it as referring only to broad gauge issues of "mission." Still others use it to denote any decision that is "important." The concept of strategy that will be explored here, and that has come to play such an important role in the study of organizational guidance, contains elements of all of the above notions. Its special focus, however, is on the relationship between a whole organization and its external environment.

Policy Hierarchies. An observable aspect of organizations is that they develop hierarchies of policy (rules for defining contexts and making choices). While one frequently speaks of policymakers as selecting courses of action, the choices are almost always policies or frameworks for guiding action. Even such a seemingly concrete decision as to "increase plant capacity by twenty-five percent within the next nine months" can also be seen as the precursor to a complex cascade of problem-solving activity. Because policy structures are hierarchical, one can speak of "top-level" policy or organizational "strategy" without meaning the combined totality of every policy and rule

in the organization.

If each level of an organization supplies structure to the task environment of lower levels, what is the framework for guiding the selection of the top-most level of policy? Unidimensional objectives that call for the maximization of profits, happiness, or the public welfare are quite useless for guiding action. Unless they are expanded into specific rules or policies for their achievement, such objectives can only function as rejection criteria. Thus the postulate of some unidimensional objective does not solve the problem of choosing top-level policy and the assumption of some set of actionable goals as "given" assumes the problem away. Somehow, the organization must maintain and make occasional changes in a set of "local" goals, or policies, that are specific enough to serve as criteria for the selection of other policies. We reserve the term "strategy" for this set of primary goals and policies.

The strategist is thus concerned with at least two levels of objectives. A substantial part of the art of strategy-making lies in discriminating among those objectives which act like values, being fixed and context free, and those which are really tools of administration, changing as strategy changes. In general, fixed objectives serve to constrain the selection of strategy, while operating objectives serve to express a strategy.

While the separation of fixed from operating objectives is a critical part of strategy formulation, it is of little concern in strategy evaluation. In appraising a strategy, the question is whether or not various objectives are consistent with one another, with the chosen context, and with policy. It does not matter where the objectives come from. Therefore, for our purposes, it will be most convenient to simply include all goals and objectives in our definition of strategy.

It is a frequent observation that one person's strategy is another's tactics - that what is strategic depends upon where you sit. One can easily find references to national, political, and industry strategies, to corporate, product, and manufacturing strategies, and even to strategies for conducting an individual interview. Common to all these problem situations is a generic type of poorly structured dilemma: a set of higherlevel objectives, frequently flowing from the resolution of a higher-level strategic problem, must be translated into local objectives and the translation depends upon both the objectives, the means at hand, and the detailed content of the situation. Without denying that strategic problems occur on many levels, our primary concern here is with organizational strategy the problem facing fairly autonomous organizations of selecting domains of activity and ways of coping with those domains.

As a descriptive tool, strategy is the analog of the biologist's method of "explaining" the structure and behavior of organisms by pointing out the functionality of each attribute in a total system (or strategy) designed to cope with or inhabit a particular niche. The normative use of strategy has no counterpart in biology (as yet!), but might be thought of as the problem of designing a living creature (or species) to exist within some environment. The hierarchical nature of policy has its analog in the fact that living systems are not homogeneous but consist of linked subsystems; the "designer," or strategist, can specify the need for "binocular vision" or "integrated production

facilities" secure in the knowledge that the problems of creating such entities are solvable.

It is again the hierarchical nature of policy that provides the justification for calling some issues "important" or "critical." Rhenman (1973) notes that most of the problems organizations face are dealt with by standard procedures they are problems that may be unpleasant, but which the system is designed to cope with. Critical or strategic problems, by contrast, lie outside the limits of the organization's repertoire of coping behavior and may require a redesign of the system. In our biological analogy, strategic issues are ones that lead to extinction unless mutation (or its behavioral equivalent) occurs. Thus, the idea of critical problems is not a capricious simplifying assumption, but has firmly grounded analogs in such concepts as "system stability ranges" and "design tolerances."

Problem Structuring. The kinds of situations that call for strategic thinking are termed ill-structured. They are characterized by difficulty, ambiguity, and the lack of established methods for moving towards clarification. According to Simon (1958):

Ill structured problems are difficult because the essential variables are not numerical at all, but symbolic or verbal. . . . the goal is vague and nonquantitative. . . . Finally, there are many practical problems - it would be accurate to say "most practical problems" for which computational algorithims simply are not available. Facing facts, we are forced to admit that the majority of decisions that executives face every day - and certainly a majority of the very most important decisions

lie much closer to the ill-structured than to the well structured end of the spectrum. To this description it is necessary to add two important sources of difficulty. First, ill-structured problems pose very real challenges in simply defining what the problem is. Even the term "ill-structured problem" tends to suggest some type of obstacle to be overcome. In reality, strategists often face situations in which there are many conflicting signs of health and disease, opportunity and risk. Arriving at a formulation of the nature of the situation may well be the principal and most crucial task. Second, the assumption of vague and nonquantitative goals is too strong to be general. In many cases, strategy-making is concerned with the creation of reasonable purposes - with moving from a general awareness of discontent to having identifiable desires.

When confronted by an ill-structured situation an individual or organization may either: (1) classify it and apply standard procedures (the bureaucratic response); or (2) seek some way of structuring the problem that is "meaningful," i.e., that suggests how current knowledge and experience can be brought to bear. The latter, more creative, response requires what Vickers (1965) calls an "appreciation" of the situation and a reduction in ambiguity through the perception of pattern. Little is known about how insight actually operates, but Schon (1963) suggests that metaphor and analogy are the substance of creative thought. New concepts arise, he contends, by the stretching and adapting of old metaphors to meet new situations. The traditional open ended question, "What business are we in?" can be seen as an invitation to metaphor generation. When Timex revolutionized the watch industry with its inexpensive disposable watches, distributed through drug

stores, it changed the prevailing metaphor from "a watch is an item of jewelry" to "a watch is a convenience item like a mechanical pencil." Similarly, when diversification away from a traditional area of business induces the senior management group to stop viewing that area as one in which new markets are "conquered," old ones "defended," distributors "strengthened," and advertising "campaigns" launched, and to see it instead as a "mature field" from which funds are "harvested" for reinvestment in more "fertile fields," a change in metaphor has taken place. The managers are, of course, neither generals nor absentee farmers, but both metaphors, though hardly "models," may provide helpful ways of organizing perceptions.

Whether or not metaphor plays a role, the primary task when facing an ill-structured problem is to reduce ambiguity by structuring the problem. Thus, for our purposes, strategymaking is a response that does not necessarily "solve" the problem, but which redefines it in terms of more familiar subproblems. When, for example, a military commander decides on a strategy of using feints to split the opposing forces into two segments, he is leaving a great many issues unresolved. He is, however, drastically reducing the ambiguity inherent in the original problem and providing a framework for further activity. Unlike rational-analytic problem solving, the function of strategy is not to "solve" a problem, but to so structure a situation that the emergent problems are solvable.

A strategy, then, is a description of what an organization is trying to accomplish and to what ends it is channeling its critical resources and problem-solving energies. As such, it may be regarded as a set of goals and policies, as a way of structuring an ill-structured situation, or as an allocation of resources. None of these views contradicts the others and all are helpful in grasping different aspects of strategy and strat-egy-making.

Strategy Evaluation Criteria

Suppose that an analyst is presented with a reasonably comprehensive description of an organization, its environment, and a proposed strategy. What criteria can and should be used in evaluating the strategy?

There may be many features of the proposed strategy that are attractive, insightful, and potentially effective. It takes, however, only one major flaw to invalidate the strategy. Thus, it is helpful to cast the evaluation problem in terms of the negative logic of hypothesis testing: on what grounds may a proposed strategy be "refuted" or rejected?

If the type of organization or environment is not specified, there are four discernible contextfree tests, each following directly from the definition and purpose of strategy, that a proposed strategy must pass: (1) the goal consistency test, (2) the frame test, (3) the competence test, and (4) the workability test.

The Goal Consistency Test. In mathematics, conflicting objectives imply a null set of feasible solutions. Similarly, a strategy that contains goals, objectives, and policies that are mutually inconsistent must be rejected. The objective of "growth," for example, is not consistent with that

of "maintaining a small informal management team."

Strategies authored by a single individual rarely suffer from such gross inconsistencies, but those that evolve from political processes often do. However, in appraising such strategies, one must realize that publicly announced goals are frequently part of the tactics for implementing an implicit strategy. President Carter's energy program, for example, contains inconsistent goals and policies: "energy conservation, promoting new energy sources, equity of sacrifice, and special taxes and penalties." Yet these statements actually represent political tactics within a larger strategy of relying heavily on legislative, rather than market, mechanisms to reduce U.S. dependence on Arab oil and soften the economic shocks forecasted to occur when oil production begins to decline.

The Frame Test. A principal function of strategy is to structure a situation - to separate the important from the unimportant and to define the critical subproblems to be dealt with. An aspect of evaluation must be an appraisal of how well these tasks have been accomplished. Before one can decide whether or not a given strategy will "work," some indication that the right issues are being worked on is needed.

The distinction between problem solving and defining the relevant problems is important in all decision situations, but especially so in strat-egy-making. It is analogous to Drucker's (1967) distinction between efficiency and effectiveness and the organizational roles of generalists and specialists. In common parlance, the ability to discriminate the important from the unimportant is usually termed "wisdom." For the sake of discussion we shall call schemes for making this kind of distinction "frame theories" to distinguish them from theories whose purpose is to specify interrelationships among known entities.

To carry out the frame test, an evaluator must have or construct a frame theory that identifies the critical issues within the chosen domain. If the strategy's basic policies and objectives do not address these issues, it must be rejected.

General frame theories, applicable to all situations, do not exist and it is doubtful if such a thing is even possible. There are, however, theories and models of varying power that serve to distinguish the important from the unimportant within specific contexts. For example, as a product/market arena evolves from the early growth to the maturity phase of its life-cycle, currently employed models predict that: (1) market share positions become entrenched; (2) cost-based competition becomes ascendant; and (3) the new stability will permit efficiency gains through vertical integration. Such statements are part of frame models because they are directed towards revealing the critical issues rather than solutions to them. Similarly, assertions that "size is vital in the automobile industry, location in retailing, and image in the liquor business," are pure frame theories (though perhaps untested) in that they indicate what one should have as an objective rather than how to accomplish it.

Many of the complaints concerning the vagueness, incompleteness, and nonoperational nature of such models stem from a basic misunderstanding of their purpose. Frame models need not, and often should not, address the question of exactly how a particular objective or requirement

should be achieved. Thus, a frame theory of coal mining might note the pivotal importance of labor productivity while a frame model of oil refining would surely ignore that issue and highlight instead the need for assured sources of supply. In each case, there are various ways in which a strategist might deal with the pivotal issue. One oil refiner might move to integrate back into tanker operations, while another, serving a relatively isolated region, might simply depend on its ability to afford a higher purchase price. Richer and more complex contexts support a greater variety of strategies, but must be framed more abstractly.

Most frame models in current use take the form of rules of thumb: focus your efforts, specialization is vital for the smaller enterprise, segment your market, etc. Some of these are of dubious generality and may apply, if at all, only to special contexts, but others represent the distillation of rather sophisticated theory. In a section that follows we shall discuss an important subset of frame models, masquerading as rules of thumb, that arise under the special context of rivalry.

The Competence Test. Since a key purpose of a strategy is to structure a situation in a way that creates solvable subproblems, strategies that do not result in solvable subproblems must be rejected. Such strategies simply substitute one ill-structured dilemma for another.

A strategy of competing in the electric typewriter market, for example, by creating a radically cheap yet durable machine defines a subproblem (inventing the machine) that is probably no more amenable to attack than the original strategic problem. The "invention" of such a device is quite unlikely unless some special resources or conditions exist. If the firm in question had a demonstrated competence at similar cost-saving innovations or held a relevant patent, the strategy might be acceptable. Thus, the idea that the subproblems defined by a strategy should be better structured than the original problem has an operational interpretation in the context of organizations - the subproblems should be ones that can be dealt with by existing and demonstrated organizational skills, resources, and competences. While it is not strictly necessary for the strategist, or strategy evaluator, to know "how" such subproblems will be solved, an appreciation for the solvability of various kinds of subproblems is required.

What is particularly interesting about this observation is that the standard rules of thumb, "build on your strengths" and "first understand your strengths and weaknesses" can be deduced from the basic proposition that strategy is problem structuring and organizational policies are hierarchical. They do not depend on assumptions about the nature of competition or the stressfulness of the environment and, consequently, apply equally to business and not-forprofit organizations. They are simply a convenient way of expressing the idea that the most direct measure of whether a subproblem is solvable, and therefore simpler than the original strategic problem, is to compare it to problems the organization has had to cope with in the past.

The Workability Test. If a strategy has a consistent goal set, passes the frame test, in that it focuses on the critical issues, and passes the competence test, in that it avoids unreasonable demands on subsequent policy-making, there still remains the question: "Will it work?" Will the proposed policies and actions work together to produce the results sought?

The issue of workability is what most formal planning documents and procedures address, and there are an enormous variety of tools, theories, and methods for making this evaluation. Most can be placed in one of two rough groups: (1) those that are concerned with the availability and deployment of resources, and (2) those that attempt to predict the outcomes associated with particular actions. The first group contains methods for financial, plant capacity, personnel, logistics, and materials flow planning. Encompassing a great deal of what is taught in formal management education, these methods range in sophistication from simple tables of cash flow to complex computer simulations and linear programming models. Their general intent is to permit proposed actions to be tested for consistency with one another and the available resources.

The second group includes procedures and models for estimating demand, price, future technology, competitive reaction, market share, etc. The best developed methods in this group are those for predicting demand. Macroeconomic models, input-output models, industry models, trend analysis, and other techniques can significantly reduce the uncertainty associated with estimates of future demand. Less well developed are models for predicting which firms, following which strategies, will satisfy this demand. Thus, while reasonably accurate forecasts of the future demand for computer peripherals are available, the rigorous prediction of the market share attainable by a specific set of price, line breadth, technical quality, and support policies is largely beyond the current state of the art. Most practical planning systems rely heavily on judgment, intuition, and debate as means of establishing these estimates. Because, except for the roughest of checks, the workability test usually requires inputs from a variety of specialized viewpoints and disciplines, it is largely carried out by a multilevel process in most organizations. (See, for example, Berg, 1965; Bower, 1970.)

This article is not an appropriate forum for surveying all, or even the most important, of the detailed methods for associating actions with results. Firmly rooted within specific functional disciplines, requiring specialized training for their use, they need to be restructured and adapted to each new problem. There are, however, a growing class of models, or theories that attempt to associate policy with results at high levels of abstraction. These shall be termed "tactical" models to stress the idea that their concern is with the manipulation of policy-type variables within a frame, rather than the choice of frame or the explosion of policy into detailed subcomponents. A model giving the direction and approximate magnitude of the associations among profitability, promotional effort, and dealer exclusivity (within, say, Porter's [1976] "nonconvenience" frame) would be classified tactical. A method for estimating segment-specific price elasticities of television sets would be classified below the tactical level of abstraction.

In general, most of the current strategy oriented research has both frame and tactical content. The builder of frame models will usually at least offer conjectures about the range of appropriate tactics within a context and the builder of tactical models is, at least implicitly, identifying the chosen parameters as critical within the given frame. Thus, as we turn to an examination of specific theories and models useful in strategy evaluation, no strenuous attempt will be made to discuss frame and tactical models in separate sections. Instead, we shall follow a general path from those

approaches that are appropriate in the widest contexts to those that are more narrowly focused. The models and theories covered include: (1) the assumption of rivalry in nonperfect markets, (2) classical evaluation approaches, (3) life cycle models, (4) empirically based frame and tactical models, (5) strategic group models, and (6) experience curve models.

Rivalry and Asymmetry: The Most General Context

The widest context that provides usable theory for strategy-making and evaluation is that provided by the simple assumption of rivalry. Strategy, as a concept, has its roots in the consideration of pure rivalry, so this should not be surprising.

Pure rivalry, as arises in games, military combat, and other interactions in which the stakes are well defined and cooperation unlikely, is a powerful simplifying assumption. Under conditions of pure rivalry, the solution, or outcome, is indeterminate under symmetry. No one can predict which of two identical opposing armies or corporations will prevail. A determinate solution only becomes possible when certain asymmetries exist, such asymmetries being termed "advantages." The winning strategy (as opposed to tactics) is always the same - play only those games in which you have an advantage. Somewhat less mechanistically, one wins games by exploiting asymmetries that make a difference. Thus, in pure rivalry, a frame model is one that tells the strategist which asymmetries constitute advantages.

One interesting consequence of pure rivalry is that the outcomes of balanced confrontations under conditions of full information will necessarily be determined by details not included in current theories of strategy. If all contenders have equal access to the available strategic knowledge, which need not be very good, the only confrontations of interest are those in which, according to the current state of strategic theory, the outcome is indeterminate. Faced with such a situation, the determined strategist will try to discover new sorts of advantages hitherto unrealized. The approach to such discovery would have to be through the study of situational asymmetry rather than the examination of theory. Once the outcome has been determined, the potential store of knowledge will have been increased - a new type of asymmetry will have been tested as to whether or not it constitutes an advantage. Therefore, in a world of pure rivalry, strategists either know in advance the outcome of a confrontation or they are pure situational thinkers. Researchers, in this imaginary world, would have to gain knowledge by the arduous and undignified method of studying a myriad of individual cases, looking for clues as to which asymmetries were critical in each.

Traditional microeconomic theory is based on the assumption of pure rivalry and frames the strategic issues quite simply: the only asymmetry that can constitute an advantage is cost. The theory goes on, of course, to make further assumptions which, in the end, guarantee equilibrium - a condition of complete symmetry. From our perspective, microeconomics is a wide context frame model with this content: (1) Cost asymmetries are critical advantages for business firms. (2) If costs are equal, the nature of advantage is beyond the scope of the theory. Of course, if an economy were

in real equilibrium, not all variables would be symmetrical, but no advantage would be attainable. In a world in which people learn, in which new insights can occur, and in which every situation contains a universe of complex potential asymmetries, it is doubtful if equilibrium is a conceivable state of affairs for anything but physical systems.

The kind of rivalry characteristic of today's economy is far from "pure rivalry." Perhaps the most important difference is its ecological nature. Not all firms are pitted against one another, protected niches exist, "mutations," in the form of innovations, constantly open up new possibilities and obsolete old sources of advantage. More important still, asymmetries that are not advantages in one arena may be critical advantages in another - we live in a world of multiple contexts. If two firms have identical costs, microeconomic theory predicts a standoff. Suppose, however, that one firm possesses a partic- ularly rich source of raw materials and the other a special manufacturing process. Each potential advantage reduces cost by the same amount so no cost asymmetry is produced. A strategist, however, might notice that the special process of the first firm gives it an advantage abroad, where it has not yet been introduced. The second firm cannot exploit the cost advantage of its raw material supply at such distances and cannot follow. It may, however, choose to investigate the use of its material advantage in other products. One firm becomes a multinational, and the other diversified. Strategy, in other words, still depends on asymmetry, but under "impure" rivalry advantages other than cost exist, information is far from perfect, and the recognition of asymmetry is as vital as its exploitation.

Rivalrous environments reward the exploitation of asymmetry and the exploitation of asymmetry, in turn, leads to the formation of niches and specialization as a means of adapting to niches. The rivalrous environment, therefore, simplifies the analytical job by usually restricting attention to the immediate niche in question and its close neighbors. Furthermore, in a particular niche, rivalry tends to be on certain key factors. Accumulated histories have given the competing organizations (or organisms) roughly equivalent endowments in the basic functions necessary for survival; factors that formed the center of strategic battles in the past (e.g., coping with labor unions, use of EDP technology) come into balance for the survivors and competition extends to new grounds.

The implications of assuming rivalry for strategy-making and the appropriate focus of strategy analysis are rarely made explicit. We have dwelt on them because statements that are frequently passed off as simply heuristics, or rules of thumb, are often deductions from a model of rivalrous competition. The model lacks the rigor so attractive to those trained in economics or the physical sciences, but is more accurate in its essentials than any available rigorous model. Biologists studying niches, ecologies, and species adaptation realize the enormous variety of solutions to competitive problems that evolving organisms can produce, and employ the same kind of situational functional analysis as do students of organizational strategy in their search for understanding. The assumption of rivalry, in sum, can be seen as a frame model that (1) invites a focus on the comparison of roughly similar entities rather than elaborate models of an entire system, (2) implies that strategic factors are discoverable through the inspection asymmetries rather than common factors, (3) implies that

stable solutions to strategic problems take the form of niches and niche maintenance behavior, and (4) implies that the fundamental threats to stability will arise from changes in the climate and innovation (mutation).

The assumption of rivalry, therefore, allows the use of a powerful rejection rule in evaluating strategy: a strategy that does not either create or exploit an asymmetry constituting an advantage must be rejected. Which asymmetries produce advantages is determined either from an appropriate frame model or situational analysis. The rule is powerful in the same way that the conservation of energy law permits one to reject proposed perpetual motion machines without studying their inner logics.

Classical Evaluation Approaches

The strategy concept as described here developed out of the contributions of a number of scholars. While Reilley (1955) and Kline (1955) were among the earliest authors to apply the term "strategy" to the idea of a comprehensive business plan, the first well articulated descriptions of strategy as a mechanism for adapting an organization to its environment were those of Selznik (1957), Chandler (1962), Tilles (1963), Learned et al. (1965), and Ansoff (1965). While Chandler saw strategy as mediating between environment and organizational structure, and Selznik saw "mission" as the essence of organizational leadership, neither author dwelt on either the formulation or evaluation of strategy. We take, therefore, Tilles's approach to evaluation, which is a distillation of the Harvard Business School concept of strategy later published by Learned et al., and Ansoff's normative prescriptions for strategy formulation and evaluation, as the "classical" approaches.

Identifying strategy as a "set of goals and major policies," Tilles suggested that strategy be evaluated on six criteria: (1) internal consistency, (2) consistency with the environment, (3) appropriateness in the light of available resources, (4) satisfactory degree of risk, (5) appropriate time horizon, and (6) adequacy of results achieved. Tilles's terminology, of course, does not match our own, but his discussion makes it clear that his first criterion corresponds to our test for goal consistency and the resource oriented part of our workability test. By "consistency with the environment" Tilles means a requirement that the strategy address and deal with the critical success factors and problems within the chosen domain - this corresponds to our frame test. The third criterion, Tilles makes clear, is a test of whether certain resources are employed to "advantage" by the strategy. Specifically mentioned are money, competence, and physical facilities. The fourth and fifth criteria are suggested elements or characteristics of the goal set and the sixth refers to a posteriori appraisal.

The Tilles aproach can be seen as a simple frame model of general business rivalry - it posits the need for some type of workability test and suggests that advantage be sought in the three broad areas of money, competence, and physical facilities. Admittedly general, the model stands in striking contrast to many prior approaches to the guidance of corporations that emphasized such factors as forecasting, superior product quality, and low costs or that suggested that success would

be attained by those who possessed "vision," "aggressiveness," or "sound management practice." While the frame model employed by Tilles is quite broad, his article recounts a number of anecdotes which serve to suggest a number of specific issues that may be critical in certain situations. Similarly, the Learned et al. text on the evaluation of strategy relies on a series of cases to make explicit the issues that the authors believe to be critical in selected situations.

Ansoff presents a normative approach to strategy formulation that emphasizes the issue of diversification. Paralleling Tilles and the Harvard group in his approach to intra-industry strategy, Ansoff suggests that diversification strategy be framed by consideration of: (1) the available funds, (2) the available level of general management skills, (3) the demand/capacity relationship in the area being considered, (4) the barriers to entry, and (5) the potential synergy. The synergy idea is the heart of the matter and, in its raw form, is simply a measure of increased efficiency through combination of activities. Ansoff, however, requires that synergy be compared to the profile of strengths and weaknesses of the firms operating in the area being considered for entry. Thus, it actually becomes a measure of the degree to which a corporate resource is capable of being an advantage in the new area. The synergy argument, therefore, amounts to the proposition that the firm should possess resources that can be used to advantage in the proposed new venture. Where Ansoff differs from Tilles and the Harvard group's approaches is over the issue of whether such advantage is required - while Ansoff values its presence highly, he does not reject the idea of diversification if synergy is absent.

In summary, these two "classical" approaches to strategy evaluation provide some very wide context frame models, but no explicit tactical theories for the evaluation process. They do provide the first comprehensive descriptions of the strategy concept, emphasize the importance of deploying resources in ways that produce relative advantage, and suggest a number of areas in which advantage will normally be found. In short, the classical approach is roughly equivalent to our basic frame of rivalry. Drawing its power from its generality, it remains the approach to which one must turn when contextspecific theories or models are not available.

Life Cycle Models

The product/market life cycle concept provides a powerful frame theory that has yet to be fully developed. Originally employed as a tool for marketing specialists, the (product) life cycle concept has been gradually elaborated to include such factors as industry structure and raw material sourcing patterns. Hofer (1975) succinctly summarizes most of the comprehensive life cycle models, including those of Levitt (1965), Fox (1973), and Wasson (1974), and also gives his own synthesis.

Most product/market life cycle models divide the evolution of a product/market area into five contexts: introduction, growth, shakeout, maturity, and decline. Ideally, if formulated as a pure frame model, a product/market life cycle model would identify those areas or issues which form the primary basis of rivalry in each context. Such a model would, for example, indicate how rivalry

in the maturity phase differs from rivalry in the growth phase, point out the issues that would be considered strategic in each phase, and highlight the areas in each phase in which asymmetry may produce advantage. Most of the product life cycle models that have been put forward, however, are mixtures of descriptions of activities, suggested actions, and frame models. Fox (1973), for example, indicates that during the growth phase rivalry will be among a few firms offering similar products, production should be centralized, and manufacturing efficiency will be the critical focus. The first point is descriptive, providing what is actually a key for identifying the phase, the second is a normative proposition (of questionable generality), and the third a frame-type statement (i.e., that advantages in efficient production will be critical) that may or may not be valid.

In most product/market life cycle models the area of greatest interest is the transition from growth to maturity, with some type of passage through a "shakeout" phase in the process. The basic idea is that the basis of rivalry and the activities of rival firms change as this transition takes place. While much of the information provided by the life cycle theorists describes "normal" or "common" activities that will occur during each phase, there is a consensus frame model, albeit broad, that is useful in strategy formulation and evaluation. Obviously pertaining only to fairly standardized products aimed at a variety of buyers, the consensus frame is approximately as follows:

In the growth phase, demand pull provides margins that may allow relatively inefficient producers to survive. Rivalry focuses on product performance, price, access to distribution channels and the identification of viable market segments and the special product charac- teristics appropriate to each. Depending on the competences of the rival firms and the nature of the market, strategies will be fashioned to take advantage of asymmetries in cost, product quality, distribution channels, and image in or access to market segments.

Shakeout is signaled by falling margins and a reduction in the rate of demand growth. By this time, market segments have been defined and footholds gained. Those with inferior products, poor distribution, high costs, or indefensible segmentation commitments have either fallen by the wayside or soon will. Rivalry shifts to a new phase - segments are cleared of weaker competitors and strategic advantage is sought in production efficiency, selling efficiency, and distribution intensity within segments. While the factors that were important during the growth phase continue to be necessary, rivals with a chance at survival already possess these skills or resources. Efficiency in one's chosen segments, and efficiencies gained by scale economies of linked segments are translated into increased ability to advertise and promote the product.

Once maturity sets in, relative stability reigns. The maturity phase, almost by definition, is one in which the rivals have established defensible territories and the attainment of critical advantage is virtually impossible. Strategy shifts to niche maintenance and the honing of established skills and tactics. Without innovation or a major change in the environment, there is no way of "winning" in a mature market unless rivals commit blunders.

As Hofer (1975) has pointed out, for the product/market life cycle frame to be truly useful in strategy evaluation, more specific contexts must be defined. The variables he suggests as important

delimiters of different contexts include product differentiability, purchase frequency, nature of the buyer's needs, rate of technological change, the ratio of distribution costs to value added, price elasticity of demand, marginal plant size, and others. Hofer and other researchers are currently attempting to build a "contingency" theory of strategy by using such variables to predict the emergent winning strategies. Hofer's fundamental frame theory is that the product/market life cycle is the most critical determinant of strategy, followed closely by the nature of buyer needs (economic vs. noneco- nomic) and the degree of product differentiation. He goes on to specify sets of resources or market characteristics that are most significant in each stage of the life cycle and offers a number of hypotheses concerning the appropriate tactics in quite specific contexts.

The contingency approach is not only a potentially fruitful source of frame and tactical theory, it also permits theory building to be directly tied to case-type data. We do, however, disagree somewhat with the idea that the models can be used to (and be validated by) successfully predict(ing) winning strategies. There is a subtle difference between trying to predict what a firm's strategy should be in a particular context and trying to identify the issues with which it must come to grips. The predictive approach, if successful, amounts to describing the common characteristics of all surviving rivals. However, among those rivals it is still asymmetry and uniqueness rather than commonality that will govern and structure the nature of current and future rivalry. Our position is that, to be useful in the formulation and evaluation of strategy, a model must not only describe what is necessary for existence in a given context, it must also indicate what the rivalry is about - the areas in which advantage will make a critical difference. How a particular firm uses its unique history and position to attain an advantage is, ultimately, beyond the reach of general theory.

A true frame model that bears a close resemblance to life cycle theories is Abernathy's model (Abernathy and Utterback, 1975; Abernathy and Townsend, 1975) on the ways in which standardization and variety act as conflicting forces to affect relative emphasis on product vs. process innovation. According to this model, the period immediately following the introduction of a complex (composed of many subunits and requiring a wide variety of materials and specialized production steps) innovative product is one in which rival firms jockey for position by testing alternative designs. During this period, constant changes in the design of the product and the wide variety of alternate forms necessary to explore the environment limit the usefulness of highly standardized production or vertical integration. Rivalry is essentially on the basis of product performance, and product innovation is the strategic issue. Eventually, the model holds, product design and the environment come into balance and a "dominant design" appears - the model T Ford, the DC-3, the vinyl LP stereo disc, etc. The consequent stability in product and reduction in variety now permit backward integration, standardization of manufacturing process, and economies of scale in integrated production to become strategic factors. While during the product innovation phase the fluidity of design did not permit a firm to firmly commit to a particular set of subassemblies or production processes, the era of stability allows production efficiency and process innovation to become critical competitive factors.

The Abernathy model also predicts that the advantages gained by standardization will eventually work against the firm if product innovation activity is renewed. Simply stated, the strategies required for success in the stable phase make the firm more and more vulnerable to the next cycle of product innovation. Of course, the reverse may also be possible - the advantage gained by the integrated producer, particularly if significant process innovation is involved, may be sufficient to ensure the economic viability of its version of the dominant design for a very long period of time. Nevertheless, the idea that strategy can be too focused, producing barriers to future adaptation in addition to barriers to entry, has direct analogs in biology and provides an intriguing counterpoint to received microeconomic equilibrium models. Some empirical support for these notions comes from the Cooper and Schendel (1976) study of how the dominant producers in seven industries responded to significant technological innovations by "outsiders." Their principal finding was that, over the long run, the traditional firms that tried to participate in the new technology were not successful.

Empirically Based Frame and Tactical Models

Cross sectional analyses of policy-performance relationships in various contexts can provide valuable insights into the underlying parameters that influence variation among firms and the range of attainable states. Such studies can produce both frame and tactical theories but remain limited by the fundamental limitation of cross sectional analysis - the observed associations do not imply causality. Two examples of fairly comprehensive cross sectional studies are the PIMS project (see Schoeffler et al., 1974; Buzzell et al., 1975) and Rumelt's (1974) study of diversification strategy.

The discovery of policy-performance relationships on a business level has historically been hampered by the unavailabiltiy of data on anything but the firm level. The PIMS project is an on-going study that is continually building a data base of business-level statistics. Administered by the Strategic Planning Institute (SPI), the project's member firms pay a fee in order to add their business-level data to the bank and, in return, gain access to the project's proprietary programs and results. SPI ensures confidentiality and supports research based on the accumulating data.

The central theme of the PIMS results is the association between market share and return on investment (ROI). This association turns out to be quite strong and is enhanced when capital intensity is low, when the product is sold to industrial customers, when product differentiation is increased, purchase frequency decreased, and in a variety of other circumstances. The published two-way contingency tables also indicate positive associations between product quality and ROI and a host of other associations. Thought provoking and suggestive, the two-way displays unfortunately leave unanswered the question of whether or not these associations continue to hold as other conditions are changed.

The PIMS data provide what is perhaps the most potent general tactical model that is available for strategy evaluation at the business level. By selecting a subsample which occupies the general

context of interest, the analyst can explore, on an ad hoc basis, the historical and cross sectional associations between variations in policy and changes in outcomes. No strong theory lies behind the observed associations, and quantitative theories concerning relationships among variables of this type simply do not exist.

In addition to providing immediate tactical tests of strategy workability, the PIMS data can be a rich source of frame theories. While much attention has been paid to the factors that influence profitability, an important empirical finding is that only a few factors explain the preponderance of profitability variance, and that these factors change as context changes. For example, while capital intensity is strongly associated with profitability among firms with moderate to low market shares, it is much less important in explaining profitability variations among firms with high market shares. It is from collections of such observations and their detailed analysis that future frame theories can be constructed.

Rumelt's (1974) study of 273 large industrial corporations focused on the relationship between profitability and diversification strategy. After defining eight patterns of diversification posture, ranging from single-business to unre-lated-portfolio, tests for long-term profitability differentials were performed. It was found that the critical explanatory factor was not the absolute degree of diversity, but the way in which businesses had been related to one another. In particular, the highest levels of profitability were associated with firms that were diversified but which related the majority of their activities to some central core skill, competence, or resource. The lowest levels of profitability were displayed by vertically integrated raw material producers and slowly growing conglomerates. Like the PIMS results, Rumelt's findings cannot be interpreted causally - it remains unclear whether the high performing group is profitable because of its strategy or whether the strategy of limited diversity reflects an unwillingness to abandon a highly profitable area. Nevertheless, the study provides this frame content: when diversification is at issue, the degree and type of interrelationships among activities tends to be a more critical decision area than the absolute degree of diversity.

In a more recent study, Rumelt (1977) has found that the high profitability of relatedconstrained firms is due to their ability to consistently take positions in industries characterized by high levels of return on capital. Furthermore, these "high-profit" industries are themselves largely composed of related-business firms.

The emergent picture is one of a class of highprofit industries existing chiefly as the intersections of related-business firms rather than as separate fields dominated by specialists. It appears that both the industry and firm phenomena are reflections of a more fundamental determinant of profitability: extensible skills together with activity in product/market arenas subject to product differentiation and market segmentation.

In summing up empirical studies on the effectiveness of strategy content, it seems evident that cross-sectional methods can be used to separate the important from the unimportant within a variety of contexts. In addition, with access to sufficient data, strong mutual associations among variables

can be used as tactical models. The fact remains, however, that without an underlying theory to explain why the observed associations occur, there are distinct limitations on these models' ability to aid one in understanding the detailed forces at work in a particular situation.

Strategic Groups Models

A branch of strategy research that promises to provide a rich source of both frame and tactical models is concerned with the discovery and understanding of heterogeneous competitive groups within industries. The traditional assumption that has guided economists in the study of industrial organization is homogeneity - that firms within an industry differ only with respect to scale. A dubious assumption at best, this approach has led inexorably to characterizing industries in terms of concentration (relative size being, after all, the only attribute the model allows) or in terms of characteristics possessed collectively by all firms in the industry (e.g., entry barriers). While the roots of an effective challenge to this tradition are many, the significant contributions currently defining the field have been made by Hatten (1974) and Patton (1976) from Purdue, strongly influenced by Schendel, Cooper, and Bass (see Bass, 1969; Hatten and Schendel, 1977; Schendel and Patton, 1976), and Hunt (1972), Newman (1973), and Porter (1976) from Harvard, strongly influ- enced by Bower and Caves (see Caves and Porter, 1977).

Hatten studied the brewing industry between 1952 and 1971 by building a linear regression model relating observable policy variables (e.g., number of plants, price, number of brands, market share, etc.) to return on equity. Using a clustering algorithm together with a statistical test for homogeneity of regression models, he showed that the industry could actually be broken into subindustries, each with its own distinct policy-performance relationship. The five clusters corresponded to large national brewers, smaller national brewers, strong regional, weak regional, and small regional brewers. Patton used the same datà but a different approach he grouped the firms into three clusters (national, large regional, and small regional) and fitted simultaneous equation regression models to each cluster with the cyclic independent variables being return on equity, market share, and efficiency (price-cost margin).

Perhaps most striking among Patton's findings is that while market share is positively and significantly related to return on equity (ROE) for the industry taken as a whole, the relationship is negative within each cluster. In other words, the national brewers have higher levels of both market share and ROE than do regional brewers, but marginal additions to market share by either group of firms lowers profitability. The regression equations thus reveal the type of niche structure that has long been the mainstay of qualitative strategic analysis.

Taken as a frame model, Patton's results suggest that rivalry among the national brewers has been essentially on efficiency and promotion exploiting economies of scale and the efficiency of new automated facilities in order to support massive promotional campaigns. The regional firms, by contrast, cannot play this game and, rather than being direct rivals, compete with the national

firms by focusing all their efforts on one or two brands in a limited, but heavy beer drinking, region.

The beer industry is one of the most mature industries and one would not expect to find either incompetent firms or hidden winning strategies that have been missed. It is, therefore, somewhat surprising to find the model suggest- ing that brand proliferation strategies and plant acquisition programs by regional brewers have generally been both unsuccessful and damaging. Of course, the model cannot tell us what would have happened to firms undertaking these programs had they not done so, nor can it reveal whether the relationship is causal or just a reflection of firms in trouble taking such actions as "last ditch" efforts. Still, the general picture presented is that efficiency on a national level is the critical issue for the large brewer, and maintaining some defensible hold on a regional area, and not expansion, is the critical issue for the smaller brewer.

Patton's model also provides a tactical model that is useful in strategy formulation and evaluation. The regression equations that have been fitted to the brewing industry are the substance of that model and provide the strategy evaluator with a description of the "strategy possibility," or feasibility, frontier. More simply, the equations provide a check on the consistency of a set of goals and policies.

Intriguing questions remain with regard to the Hatten and Patton studies. How unique is the beer industry? Why have so few beer firms diversified to any extent? Is the observed pattern generalizable or a function of the rather unique characteristics of the beer market (e.g., unusually high resistance to new brands)?

Porter's investigation of bilateral market power provides a new frame model and additional support for the strategic group concept. Focusing on the interaction between manufacturers and retailers, Porter hypothesized that the manufacturer's profits would, other things being equal, decline as retailer bargaining power increased. That power, he reasoned, depends chiefly on the retailer's ability to help differentiate the product. Splitting retailers (and products) into "convenience" (frequently purchased goods sold through self-service outlets) and "nonconvenience" (shopping goods sold through specialty stores and/or dealers) outlets, the latter would be expected to have a greater impact on product differentiation. Manufacturers' primary weapon in countering retailer power is strongly advertised brands since stores can hardly deny shelf space to items that consumers demand. Using these and other arguments, Por- ter built a model predicting that manufacturer profit rates would vary strongly with advertising intensity in convenience goods, but that advertising intensity would have a much smaller influence on the profitability of makers of nonconvenience goods. The prediction was strongly supported by several tests comparing the form of traditional industry structure models in each of the two contexts.

Porter states quite clearly that his intention was to build and test a frame-type model (1976: 234): "The model identifies what strategic elements are crucial in determining profits and which ones have little effect." His results not only suggest that strong brands are the critical issue in convenience goods, but that there is little room for strategic variability in convenience goods. A simple bilateral model explains a great deal of the profitability variance. By contrast, branding and

advertising are less critical issues in nonconvenience goods and the room for strategic variability is much greater. Thus, while there are only a few viable strategies in the convenience goods arena, nonconvenience goods businesses can seek advantage in many different ways (e.g., branding, service, product quality, geographical specialization, type of outlet, and so on). Porter has not only provided a new model, he has significantly increased our ability to speak clearly about and validate frame-type models.

The Bcg Experience Effect

The Boston Consulting Group (1968) has taken the well-known learning curve effect, generalized it to apply to whole business units, and built a fairly elaborate structure that is essentially a strongly quantitative tactical model. According to BCG, the cost elements of value added tend to drop by 20 to 30% each time accumulated production (in units) doubles. In essence, BCG is arguing that the general observable trend of decreasing costs (adjusted for inflation) is due to a special type of scaleeconomy that must be earned by experience rather than purchased. The immediate consequences of this proposition are: (1) of firms following roughly parallel expansion paths, the one with the largest market share will have the lowest unit costs; (2) as long as the product does not change radically, and providing that the market leader commits no blunders, trailing firms will never be able to overtake the market leader; (3) accumulated experience, or its proxy, market share, is a valuable resource and rational strat-egy-making requires that it be treated as an investment; and (4) eventual market dominance is determined early in the struggle for position.

Economics is supposed to be the "dismal" science; the BCG theory is dismal news indeed for many strategists. It suggests that most market positions currently being worked on are hopeless, that taking high profits in a growth industry is the road to disaster, and that the game, in the end, goes to the swiftest and most efficient.

The basic model's frame content is straightforward. The areas in which advantage occur are in having a good product ready to market earlier than one's rivals and in having the financial and organizational strength to sustain more rapid growth (at perhaps a lower margin!) than one's rivals. Like all frame models, this one is silent on the issues of gaining these advantages in the first place. BCG tends to suggest that a properly managed diversified firm has the best mix of resources to provide these advantages. In theory this may be true, but few diversified firms have been able to solve the problems of getting innovative new products to market early and of supporting division-level strategies of foregoing current profit for future return.

The BCG framework is also a powerful tactical model, especially when combined with some model of price elasticity of demand. The analyst can, by setting a few key parameters, estimate the cash flow accruing to the firm under a variety of pricing and growth policies.

The major stumbling block in using the BCG model is that the contexts in which it is valid are not well established. The electronics industry, among others, appears to be a context which

supports the model, but it is not clear to what degree it applies in areas characterized by heavy branding and promotion, highly segmented markets, custom engineered products, etc. In addition, anyone who has tried to tie down the concept of "market share" in any but the simplest industries is aware how vague a concept it really is.

Our conjecture is that the BCG model will continue to be elaborated and the contexts within which it is appropriate will become better defined. In addition, it seems probable that the concept of "shared experience" among different products will ultimately prove to be a more fertile ground for strategic models than the raw experience effect theory. For example, current strategic theories almost invariably imply that persistently profitless business areas be dropped. The idea of shared experience, by contrast, suggests that careful attention must be paid to the possibility of continuing to gain experience in a low profit area in order to lower costs in a related, but more profitable, business.

Summary and Suggested Directions for Further Research

Focusing on analytical approaches, we began our examination of strategy evaluation by noting that the primary function of strategy is to provide a consistent set of objectives and policies that restructures an ambiguous reality into a set of organizationally solvable subproblems. We then argued that this purpose implied four tests that must be passed if a strategy is to be judged acceptable. They were:

1. The Goal Consistency Test-a proposed strategy must be rejected if it contains mutually inconsistent goals, objectives, and/or policies.

2. The Frame Test - the strategy must be rejected if it does not address those issues or problem areas which are crucially important within the chosen context. Assuming rivalry, the strategy must exploit asymmetries which constitute advantages within the chosen domain.

3. The Competence Test - the strategy must be rejected if the subproblems it poses do not lie within the realm of those that are solvable by demonstrated resources or organizational skills and competences.

4. The Workability Test - the strategy must be rejected if the combined policies are infeasible from a resource point of view or if available knowledge indicates that the desired objectives will not be obtained by the policy set.

These tests are context free and apply to any organization in any environment. Turning to the problem of making these tests operational, we examined, in progressively narrower contexts, the frame and tactical aspects of a number of recent research efforts and contributions to theory building pertaining to the content of effective strategies.

What, then, is the state of the art? Our survey of approaches was illustrative but necessarily incomplete. Omitted was the vast literature on industrial organization, a goodly number of specific tactical models, many insightful, but as yet untested, normative propositions, the whole political process approach to the strategy formulation/evaluating issue, and many other worthwhile contributions. Nevertheless, even if our survey had been comprehensive, the simple answer would have to be that strategy evaluation is still pretty much of an art. Progress is being made, but that progress is itself an art practiced by individuals with widely varying points of view, employing disparate techniques, and aiming toward different goals. Knowledge is accumulating, but there has not, as yet, appeared a paradigm powerful enough to permit its integration. Strategy continues to be studied because of its importance rather than its tractability.

A more complete answer would have to begin by noting that the approaches examined in this article appear to conform to a pernicious variant of Heisenberg's uncertainty principle: apparently, one can gain concreteness only through a narrowing of scope, and breadth of vision is purchased at the price of essential detail. The classical approach permits the consideration of the broadest range of issues, but provides only the flimsiest structure and no calculus for the manipulation of the constructs it suggests. The BCG model, by contrast, attains a crystal clear inner logic by focusing almost entirely on the single issue of unit cost.

Further Directions. Accepting the need for work to continue on several levels of abstraction, some of the crucial questions that future research should probably address if a coherent body of knowledge is going to evolve are the following:

1. Much better models and theories are needed of how firms actually compete. How do book publishers and motion picture producers compete? What is the nature of the rivalry among aerospace contractors and among office equipment manufacturers? Microeconomics is essentially a system of logic which, once one understands what contributes to "cost" in a given situation, can always provide an a posteriori explanation. What is needed is a simplified model of rivalry that still captures its multidimensional nature.

2. Manufacturing has been woefully neglected as an aspect of strategy. Scherer's (1975) study of the economies of multiplant operation provides more insight into the policy issues surrounding manufacturing than any "policy" text.

3. Studies that use profitability as the dependent variable tread very thin ice. If we believe our concepts concerning adaptation to niches, specialization, segmentation, and rivalry based on asymmetry, comparing profitabilities across niches (or even strategies) tells us very little about "success."

4. Statistical tests of theories need to be supplemented with tests of their efficacy in aiding understanding of specific strategic situations. McDonald's (1975) game theoretic analysis of

the Ford-GM confrontation shows what can be done with the simplest tools. If our theories are more sophisticated, cannot we do more?

5. We need an improved understanding of the determinants of industry structure and a better language for describing the structure of industry. The strategic groups concept represents a beginning of the move away from equating structure with concentration, but much more work is required.

6. Methodologies for testing the validity of frame statements and frame models need to be created and legitimized. Porter's (1976) study provides an important example of how this can be done.

Turning to a more specific set of research opportunities, the life cycle models focus on the issue of adaptation to change, and particularly change that is induced by the rivalry itself rather than from external sources, but need to be more sharply focused on the nature of rivalry in each phase. By contrast, the Hatten-Patton type of model reveals the inner structure of rivalry in sharply quantitative terms, but has been applied so far only to an atypically stable situation. What might be fruitful would be a study that applied the Hatten-Patton method to a competitive group in the growth phase and then again in the maturity phase of the life cycle. Several of the key hypotheses of the life cycle theory might be tested by such an approach.

The PIMS data base will obviously provide fertile territory for a number of studies. In addition to supporting further tests of the life cycle theories, it should also provide the means to test a broad range of frame models. Perhaps most intriguing will be research aimed at explaining some of the counter-intuitive results already reported by PIMS. The negative association between R and D intensity and profitability has already been the subject of a prize-winning thesis by Christensen (1977).

The Boston Consulting Group's model will undoubtedly spawn a variety of operations research oriented elaborations, but its real utility in strategy formulation and evaluation will only be realized if the contexts within which it is valid are better established. A practical place to begin examining this issue might be an identification of the factors which influence the degree to which experience is a firm versus industry effect. It may well be that such factors are the real elements of strategy in situations where the experience effect is strong.

A Closing Note. In addition to its obvious value to practicing managers, the major opportunity that beckons in the study of strategy and strategy evaluation is a significant restruc- turing of the theory of the firm and the theory of industrial structure. Although this opportunity is, perhaps, more a function of the slowness of professional economists to come to grips with these issues than of great strides on the part of students of corporate strategy, it nonetheless exists. Economic studies of competitive behavior have traditionally been oriented towards a particular set of audiences: other economists, Congress, the Federal regulatory agencies, the Department of Justice, and the courts. The theories developed have generally been designed to help determine whether or not markets

are "competitive," to evaluate certain practices as to fairness, and are heavily influenced by the axiomatic theory of welfare economics.

What the economics profession has not been able to deal with are precisely those issues that are central to strategic theory: the rivalry of firms, the inner details of industries, the dynamics of competition based on innovation, product differentiation, segmentation, and the shifting defense and exploitation of changing territories. These phenomena have not been ignored because they are invisible, but because the bulk of traditional economic models assume away such behavior. Good economists talk about these issues, using the symbol system of economics, and are fully aware of their importance. The overwhelming weight of traditional theory, however, tends to relegate such observations to the status of folklore - they are fascinating, but not suitable subjects for rigorous theorizing.

This opportunity is a significant one because it holds out the promise that public policy may ultimately be influenced by a more valid view of the functioning of a modern economy.

Commentary

This comment on Rumelt's paper focuses on three major points: strategy definitions, evaluation criteria and a comment on his literature review of strategy evaluation research. Then, Rumelt's thoughts are placed in the broader context of strategy as a totality: formulation, content and implementation, with some remarks as to where it all points.

Strategy Definition

Rumelt begins by defining strategy, selecting observations on the subject from many sources and correctly noting that the strategy concept can be applied to activities at all levels of an organization. But rather than weave an hierarchical explanation upon which to base the rest of his remarks, Rumelt leaves the reader with an interesting array of possible choices, but no clear understanding of what he means by strategy. Rumelt apparently considers strategy to be top level managers' goals and policies that structure an otherwise ambiguous organizationenvironment interrelationship. Unfortunately, within this definition, Rumelt fails to separate the formulation process of those goals and policies from the "...set of goals and policies..." that emerge from the process. It is wise, however, to separate process from content for evaluation purposes, for, contrary to Rumelt's opening statement, all aspects of ill-structured problem solving offer outstanding opportunities for rational analysis.

Evaluation Criteria

Rumelt suggests four context-free tests by which goals and policies may be evaluated: (1) goal consistency, (2) frame, (3) competence, and (4) workability. Goal consistency, competence, and workability are concepts familiar to strategy evaluators, but the frame test as Rumelt presents it is unique in the strategy literature. According to Rumelt, frame theories are schemes for discriminating important from unimportant issues within unstructured situations. He acknowledges that "general frame theories applicable to all situations do not exist..." and likens them to "rules of thumb" by which "critical issues" are identified. This definition implies that frame theories can be used in the strategy formulation process. Thus, frame theories are not solely tools of evaluators, but also of formulators, a confusion which is endemic in theoretical efforts to separate evaluation from formulation.

Foremost among frame models, Rumelt says, is the model of "rivalry." His discourse on the theory of rivalry makes inaccurate references to micro-economic theory and fails to provide convincing logic that rivalry is more accurate or different from micro-economic theories of competition. More importantly, he correctly states that under conditions of competition, strategies must create or exploit advantages. He is correct in noting that micro-economic theories inadequately express the complexities of strategies. But this is primarily because of their unrealistic assumptions, especially the assumption of maximum profits as the sole goal of business firms.

Literature Review of Evaluation Research

Rumelt provides a thorough and useful, organized review of the available literature on techniques of evaluating organizational goals and policies. He attempts to apply his unique concept of "frame models" as a link to connect the otherwise diverse literature. Frame models, however, do not fit all the literature, so application of the concept requires variation in its meaning. For example, Tilles' six evaluation criteria are not tools for separating the important from the unimportant as is required by Rumelt's definition of a frame model. Tilles' tests of (1) internal consistency, (2) environmental, (3) resource appropriateness, (4) degree of risk, (5) appropriate time horizon, and (6) adequacy of results are applied after the critical issues have been identified. On the other hand, the (product) life cycle theory fits the frame model concept as Rumelt suggests.

In his search for frame models within the literature, Rumelt overlooks a significant empirical research contribution. He cites Patton's (1976) dissertation and notes its importance for "strategic group models." He overlooks Patton's most important contribution however, which is the use of multiple equations to express the true context of multiple goal seeking organizations. Micro-economists have unrealistically assumed away multiple goal strategies, choosing instead to maximize profit. Strategy researchers have, by and large, followed this lead by choosing ROI as the measure of overall firm success. However, Patton breaks with this tradition, using the rigor of statistical analysis

to create multiple "goal" equations.

A Broader Context

Strategy evaluation is as broad as Rumelt suggests, perhaps broader. Thus, one should examine aspects of managerial behavior along with evaluation of content of goals and policies when evaluating strategy.

By defining strategic management to mean the overall managerial process for establishing strategy (goals and policies) and guiding the organization's actions towards their attainment, the broader context of strategy evaluation becomes more evident. In examining this broader context, evaluation can be divided into three parts: (1) evaluation of the strategy formulation process, (2) evaluation of strategy content, and (3) an evaluation of strategy implementation capability.

Evaluation of the Strategy Formulation Process

Strategy formulation is a behavioral process, involving thought patterns and behaviors of individuals and groups as they perceive and process information in their efforts to structure unstructured situations. As a behavioral process, it requires behavioral evaluation tools. Decision behavior research is a broad field and no comprehensive compendium of empirical findings exists. Projections of future research directions, however, have been offered by activists in this field. Noteworthy among these are Mintzberg (1973) with his unique insights, McClelland (1967) who has identified power, affiliation and achievement motivation among managers, and Van de Ven (1975) who has explored group planning processes. [1] [1] Editor's note: A very profitable direction for future policy research would be to integrate some of these findings into the models used by practitioners to evaluate the effectiveness of the strategy formulation process.

Strategy Content Evaluation

Current empirical research thrusts in strategy content evaluation are directed towards converting subjective tools into analytical models. The focus of these efforts is relating organization performance to strategy through the underlying assumption that performance is a result of organization actions intended to achieve objectives, subject to certain conditions.

PIMS (Buzzell, et al., 1975) researchers, Hatten (1974), Kirchhoff (1975a), and Patton (1976) use multiple regression with results that suggest the methodology is appropriate. These models, though, suffer one or both of two limitations: (1) their narrow perspective on performance (must use only ROI as a performance measure), and (2) their exclusion of organization aspirations as evaluation criteria. As already emphasized, Patton's multiple equation model is free of the first limitation, but subject to the second.

The problem of incorporating organization aspiration (goals and objectives) in strategy evaluation is complex. If organization aspirations are used to guide actions, then the actual results of these actions should be measured relative to the desired results. This forms a rigorous test of goal content feasibility. Researchers should develop methods of operationalizing this concept. For example, many organizations have formal planning systems, so written records of goals should exist. Based on such records, the ratio of actual ROI divided by goal ROI could be substituted for raw ROI as a performance measure.

Evaluation of Strategy Implementation Capability

Analytic models built upon examinations of performance results assume that organizations can act so as to cause performance results. By implication, the researcher assumes that organization goals and strategy are directly translated into actions. This assumption is drawn from the classical model of bureaucracy, i.e., top management sets goals and organization members pursue them. But organization theory (March and Simon, 1958; Perrow, 1961) and empirical research (Likert, 1967; Stogdill, 1974; Fiedler, 1967) question this assumption. Thus, strategy implementation capability cannot be assumed and must be evaluated along with content and formulation capability.

Evaluation of implementation capability should focus on whether managers at top, middle and lower levels actually connect their day-to-day work to the organization's goals. This is again a behavioral research problem. A review of empirical research on leadership reveals an immense number of laboratory and field research studies bearing on this question, but this research draws data primarily from lower level managers, workers, and other nonmanagers, and measures performance in terms more tactical than strategic, e.g., job satisfaction, role conflict, absenteeism, etc., and in terms that do not deal with overall organization performance.

A review of the leadership literature gives some clues to useful research on the evaluation of implementation capability, but provides few appropriate instruments for assessing how top level managers pursue goals and generally use strategy to guide operations. For example, Kirchhoff's (1971) survey research work found upper level managers occasionally refused to respond on Likert's (1967) "Profile of Organization Characteristics" instrument and Fiedler's (1967) "Least Preferred Co-worker" instrument. Post test interviews revealed that these top level managers felt intimidated and/or alienated by the concepts and vocabulary of these instruments.

Because of these difficulties, Kirchhoff (1975b) created a measuring instrument using vocabulary and conceptual schemes familiar to upper level managers. His "Managerial Style Questionnaire" (MSQ) has had a high response rate, even in mail surveys. It measures the extent to which managers use objectives and goals in performing their managerial duties. Based upon responses of middle and top level managers in twelve organizations, the MSQ satisfactorily meets psychometric standards for validity and reliability. Research results so far indicate: (1) that the use of objectives varies from organization to organization; (2) that existence of formal written objectives is positively

related to the use of objectives, but does not guarantee their use; and (3) that managers can be trained to use objectives. In addition, the use of objectives by organizations has been found to be positively correlated with overall performance within 31 profit centers of a large U.S. manufacturing corporation (Kirchhoff and Schendel, 1977).

The need for better criteria to evaluate implementation capability poses a challenge to behaviorally-trained researchers to define new, broad behavioral constructs and to operationalize them for field research with top level managers. Strategists need to know more about how goals and strategy translate into actions, and what kinds of managers within different organizations effectively carry out actions intended to reach goals. The use of such new behavioral constructs can help improve our knowledge of these processes.

Summary

The evaluation of implementation capacity brings us full circle back to strategy formulation, for many implementors are also formulators. In reality, the formulation, content and implementation of strategy are inseparable parts of the strategic management process. Conceptually, however, they must be separated to differentiate evaluation methods and direct future research. At this stage of development of strategy evaluation research the real challenge for strategy evaluation researchers is to incorporate economics, sociology, psychology and working manager pragmatism into a comprehensive model for evaluation of strategy and performance. The pursuit is multi-disciplinary.

Section 4 References

Abernathy, William J. and Phillip L. Townsend. "Technology, Productivity, and Process Change." Technological Forecasting and Social Change, Vol. 7, No. 4, 1975.

Abernathy, William J. and James M. Utterback. "Innovation and the Evolving Str

Andrews, Kennth. The Concept of Corporate Strategy. Homewood, Ill.: Dow-Jones-Irwin, 1971.

Ansoff, H. Igor. Corporate Strategy: An Analytical Approach to Business Policy for Growth and Expansion. New York: McGraw-Hill, 1965.

Berg, Norman A. "Strategic Planning in Conglomerate Companies." Harvard Business Review, May-June 1965.

Boston Consulting Group Staff. Perspectives on Experience. Boston: The Boston Consulting Group, 1968.

Bower, Joseph L. "Planning within the Firm." American Economic Review, May 1970.

Braybrooke, David and Charles E. Lindblom. A Strategy of Decision. New York: The Free Press, 1963.

Buzzell, Robert D., Bradley T. Gale and Ralph G. M. Sultan. "Market Share - Key to Profitability." Harvard Business Review. January-February 1975.

Caves, R. E. and M. E. Porter. "From Entry Barriers to Mobility Barriers: Conjectural Decisions and Contrived Deterrence to New Competition." Quarterly Journal of Economics, May 1977.

Chandler, Alfred D. Strategy and Structure: Chapters in the History of the American Industrial Enterprise. Cambridge, Mass.: M.I.T. Press, 1962.

Christensen, H. Kurt. "Product, Market, and Company Influences upon the Profitability of Business Unit Research and Development Expenditures." Doctoral diss., Columbia University, 1977.

Cooper, Arnold C. and Dan Schendel. "Strategic Responses to Technological Threats." Business Horizons, February 1976.

Dhalla, Nariman K. and Sonia Yuspeh, "Forget the Product Life Cycle Concept!" Harvard Business Review, January-February 1976.

Drucker, Peter F. "The Effective Decision." Harvard Business Review, January-February 1967.

Fiedler, Fred E. A Theory of Leadership Effectiveness. New York: McGraw-Hill, 1967.

Fox, Harold. "A Framework for Functional Coordi- nation." Atlanta Economic Review, NovemberDecember 1973.

Gale, Bradley T. "Market Share and Rate of Return." Review of Economics and Statistics, November 1972.

Hatten, Kenneth J. "Strategic Models in the Brewing Industry." Doctoral diss., Purdue University, 1974.

Hofer, Charles W. "Toward a Contingency Theory of Business Strategy." Academy of Management Journal, December 1975.

Hofer, Charles W. "Research on Strategic Planning: A Survey of Past Studies and Suggestions for Future Efforts." Journal of Economics and Business, Spring-Summer 1976.

Hofer, Charles W. and Dan Schendel. Strategy Formulation: Analytical Concepts. St. Paul, Minn.: West Publishing Company, 1978.

Hunt, Michael S. "Competition in the Major Home Appliance Industry, 1960-1970." Doctoral diss., Harvard University, 1972.

Kirchhoff, Bruce A. A Foundation for the Measurement of Management by Objectives. Salt Lake City: Bureau of Economic and Business Research, University of Utah, 1971.

Kirchhoff, Bruce A. "Empirical Analysis of Strategic Factors Contributing to Return on Investment." Academy of Management Proceedings, August 10, 1975a.

Kirchhoff, Bruce A. "A Diagnostic Tool for Management by Objectives." Personnel Psychology, Autumn 1975b.

Kirchhoff, Bruce A. and Dan Schendel. "Linking Strategy and Operations through MBO." Presented at the Academy of Management National Meetings, Orlando, Florida, August 1977.

Kline, C. H. "The Strategy of Product Policy." Harvard Business Review, July-August 1955.

Learned, Edmund P., C. Roland Christensen, Kenneth R. Andrews and William D. Guth. Business Policy: Text and Cases. Homewood, Ill.: Richard D. Irwin, 1965.

Levitt, Theodore. "Exploit the Product Life Cycle." Harvard Business Review, November-December

1965.

Likert, Rensis. The Human Organization: Its Management and Value. New York: McGraw-Hill, 1967.

March, James G. and Herbert A. Simon. Organizations. New York: Wiley, 1958.

McClelland, David C. The Achieving Society. New York: The Free Press, 1967.

McDonald, John. The Game of Business. Garden City, N.Y.: Doubleday and Co., 1975.

Mintzberg, Henry. The Nature of Managerial Work. New York: Harper and Row, 1973.

Newman, Howard H. "Strategic Groups and the Structure-Performance Relationship." Doctoral diss., Harvard University, 1973.

Patton, G. Richard. "A Simultaneous Equation Model of Corporate Strategy: The Case of the U.S. Brewing Industry." Doctoral diss., Purdue University, 1976.

Perrow, Charles. "The Analysis of Goals in Complex Organizations." American Sociological Review, December 1961.

Porter, Michael E. Interbrand Choice, Strategy, and Bilateral Market Power. Harvard Economic Studies, Vol. 146. Cambridge, Mass.: Harvard University Press, 1976.

Reilley, E. W. "Planning the Future Strategy of the Business." Advanced Management, December 1955.

Rhenman, Eric. Organization Theory for Long Range Planning. New York: Wiley, 1973.

Rumelt, Richard P. Strategy, Structure, and Economic Performance. Boston: Division of Research, Graduate School of Business Administration, Harvard University, 1974.

Rumelt, Richard P. "Diversity and Profitability." Managerial Studies Center Working Paper MGL51, Graduate School of Management, University of California at Los Angeles, 1977.

Schendel, Dan E. and G. Richard Patton. "A Simultaneous Equation Model of Corporate Strategy."

Scherer, Frederic M. The Economics of Multi-Plant Operation: An International Comparison Study. Cambridge, Mass.: Harvard University Press, 1975.

Schoeffler, Sidney, Robert D. Buzzell and Donald F. Heany. "Impact of Strategic Planning on Profit Performance." Harvard Business Review, MarchApril 1974.

Schon, Donald A. Displacement of Concepts. London: Tavistock Publications, 1963.

Selznik, Phillip. Leadership in Administration. New York: Harper and Row, 1957.

Simon, Herbert A. "Decision Rules for Production and Inventory Controls with Probability Forecasts of Sales." O.N.R. Research Memorandum, Carnegie Institute of Technology, 1958. Quoted in Shull, F. A., A. L. Delbecq and L. L. Cummings. Organizational Decision Making. New York: McGraw-Hill, 1970.

Stodgill, Ralph M. Handbook of Leadership. New York: The Free Press, 1974.

Tilles, Seymour. "How to Evaluate Corporate Strategy." Harvard Business Review, July-August 1963.

Van de Ven, Andrew H. Group Decision Making and Effectiveness: An Experimental Study. Kent, Ohio: Administrative Research Institute, Kent State University, 1975.

Vickers, Sir Geoffery. The Art of Judgment: A Study of Policy Making. New York: Basic Books, 1965.

Wasson, Chester R. Dynamic Competitive Strategy and Product Life Cycles. St. Charles, Ill.: Challenge Books, 1974.